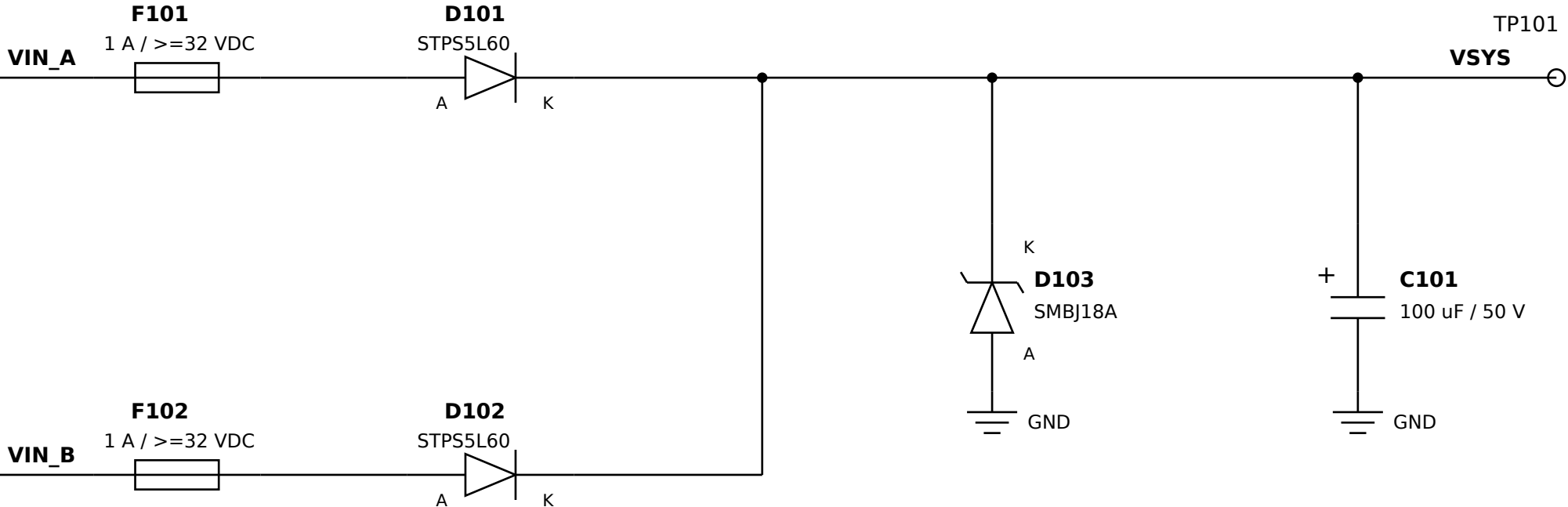
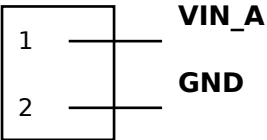


STRIJDER VISION / CB-01

01 Dual-source low-power entry

12-16 V DC laboratory supplies only; 1 A combined board-input ceiling. Optional backup source is externally managed.

J101 MAIN DC INPUT



F101/F102 must be source-adjacent inline fuses; they protect wiring before this board.

Diode cathodes (banded ends) join VSYS. This is higher-voltage-wins OR-ing, NOT main-source priority.

Common GND is intentional. No galvanic power isolation, battery charger, BMS or undervoltage cutoff is provided.

D103 is a small transient clamp, not an automotive load-dump solution or sustained-overvoltage disconnect.

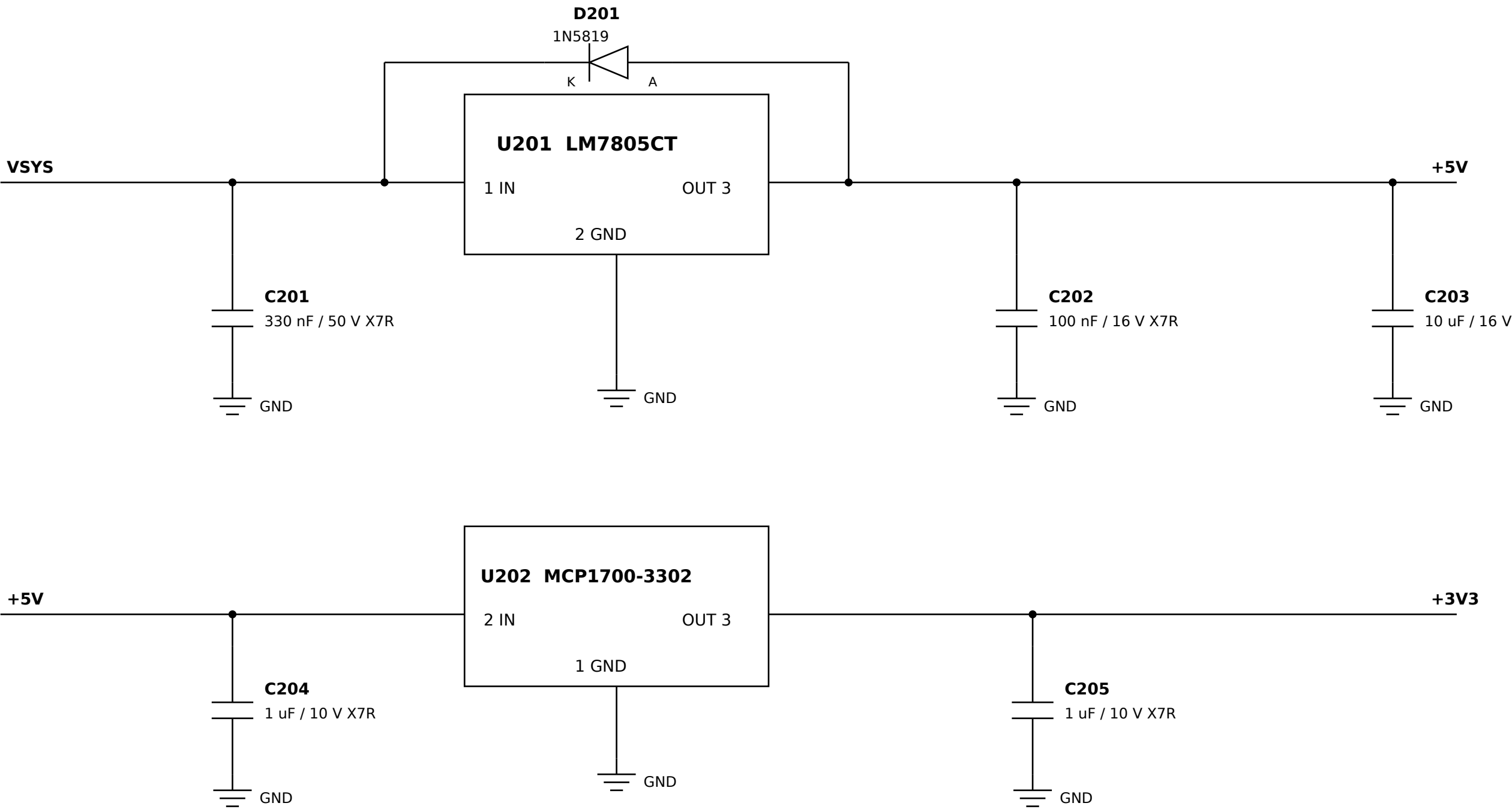
Do not connect directly to a vehicle or an unprotected battery. Do not parallel independent grounded supplies blindly.

Reference parts: ST STPS5L60; Littelfuse SMBJ18A. VSYS is unregulated and lower than the winning input.

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02 Low-power 5 V and 3.3 V rails

Bench control board only. Combined 5 V regulator load <=200 mA, including all downstream 3.3 V current.



3.3 V total budget: 50 mA. External MCU must be self-powered and use a common GND; do not tie its supply to these rails.

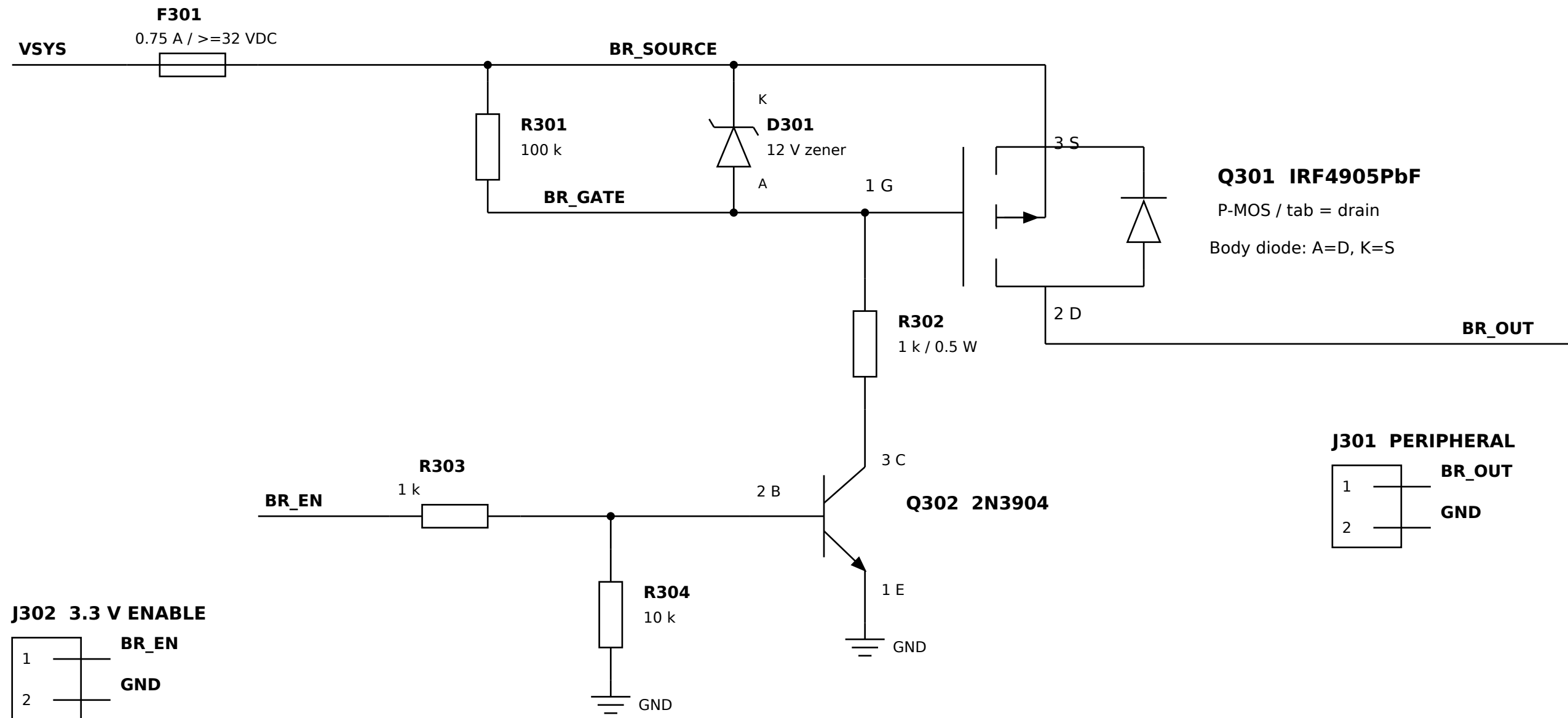
U201: fit heatsink <=15 K/W sink-to-air; allow <=6 K/W junction/interface. Verify case temperature at 16 V input.

Conservative U201 heat at VSYS=16 V, Iout=0.20 A: 2.20 W plus ground-current loss. Not a compute power supply.

D201: anode on +5V, cathode on VSYS. Capacitors close to pins. No external power injection on +5V or +3V3.

03 Switched peripheral DC output

One 0.5 A resistive/low-capacitance bench channel. No PWM, inductive loads, lightbars or vehicle actuator connection.

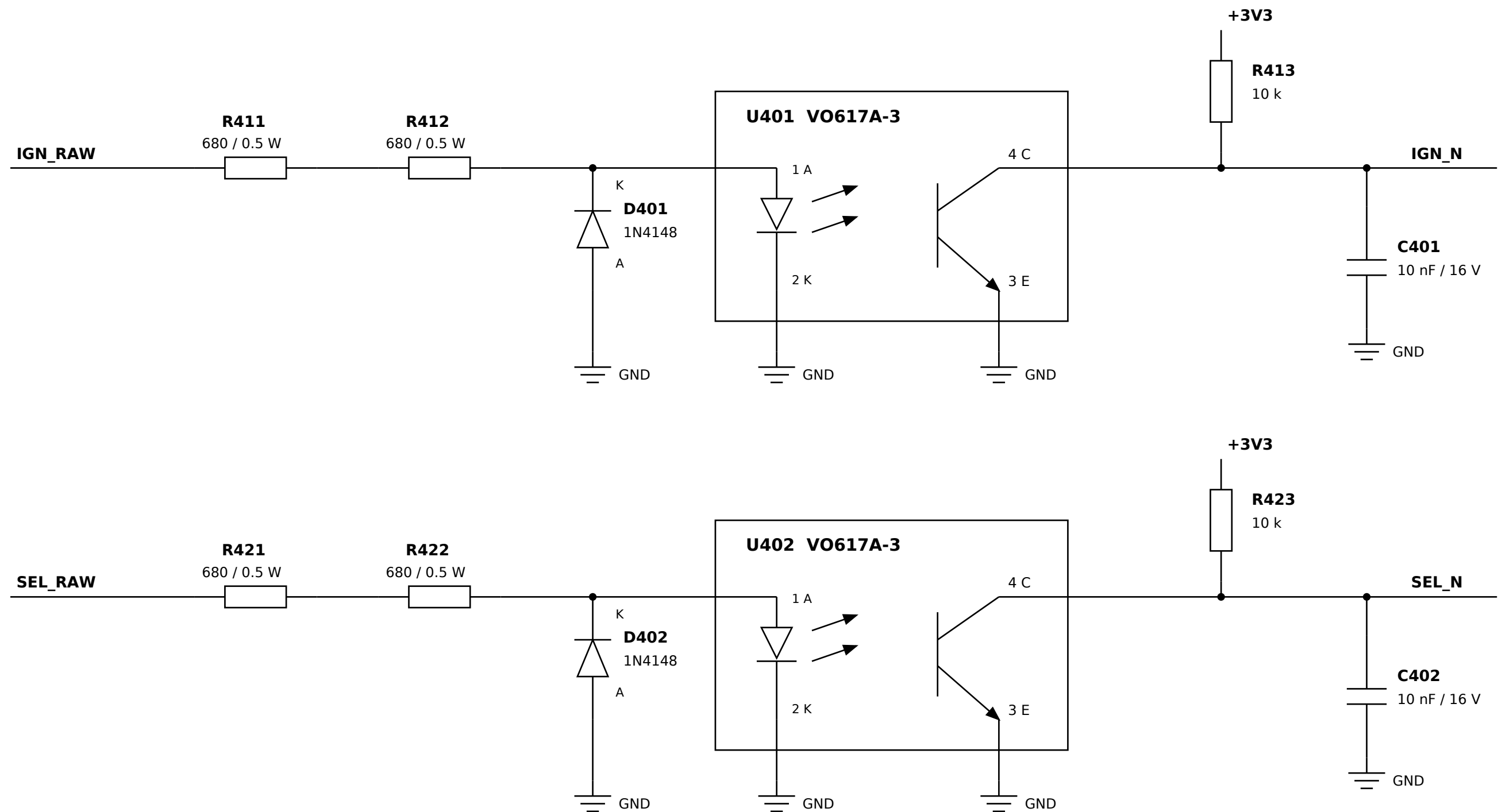


Default OFF: R304 holds Q302 off; R301 returns Q301 gate to source. BR_EN=3.3 V enables the branch.
D301 cathode at SOURCE, anode at GATE. R302 limits zener/collector current. Verify VGS at both input extremes.
F301 is a wire-protection fuse, NOT a 0.5 A electronic current limit. Short-circuit survival is not established.
Use a current-limited supply and resistive dummy load first. External output power backfeeds through Q301 body diode.
No reverse blocking, current telemetry, UVLO, controlled inrush or independent watchdog is implemented in this revision.

STRIJDER VISION / CB-01

04 Ignition and selector input conditioning

Two populated active-high 9-16 V sense channels. Output is active LOW. Common return means no system isolation claim.



Connector J401: 1=IGN_RAW, 2=SEL_RAW, 3=GND. J402: 1=IGN_N, 2=SEL_N, 3=GND (to 3.3 V MCU inputs).

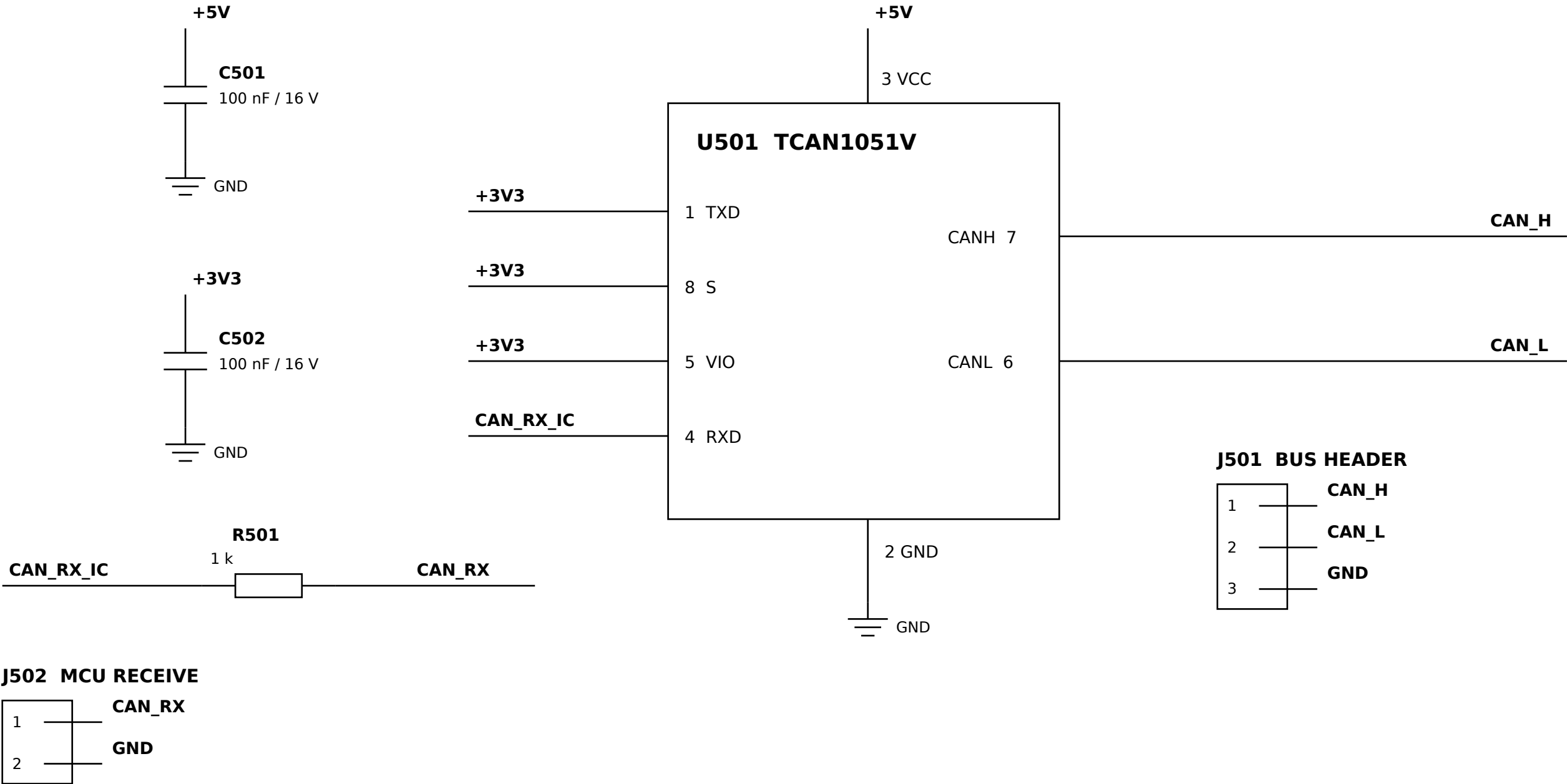
Both connectors are logical bench assignments. Detect 9-16 V asserted; 0-1 V absent; intermediate voltages undefined.

Specify Schmitt-trigger MCU inputs and ≥ 20 ms software debounce. These circuits do not implement precision thresholds.

STRIJDER VISION / CB-01

05 Hardware-silent CAN receiver

TCAN1051V-Q1 SOIC-8 only. S and TXD tied HIGH; no MCU transmit conductor and no fitted bus termination.



C501 at VCC pin 3, C502 at VIO pin 5; short return to pin 2. Do not substitute the non-VIO TCAN1051 variant.

No TVS/filter is fitted in this base: short-cable terminated bench bus only. Vehicle ESD/EMC protection remains open.

MCU CAN controller must also use listen-only mode. S is permanently high in copper; firmware cannot enable TX.

External bench bus must already have its two correct endpoint terminations. Do not add a third 120-ohm load.